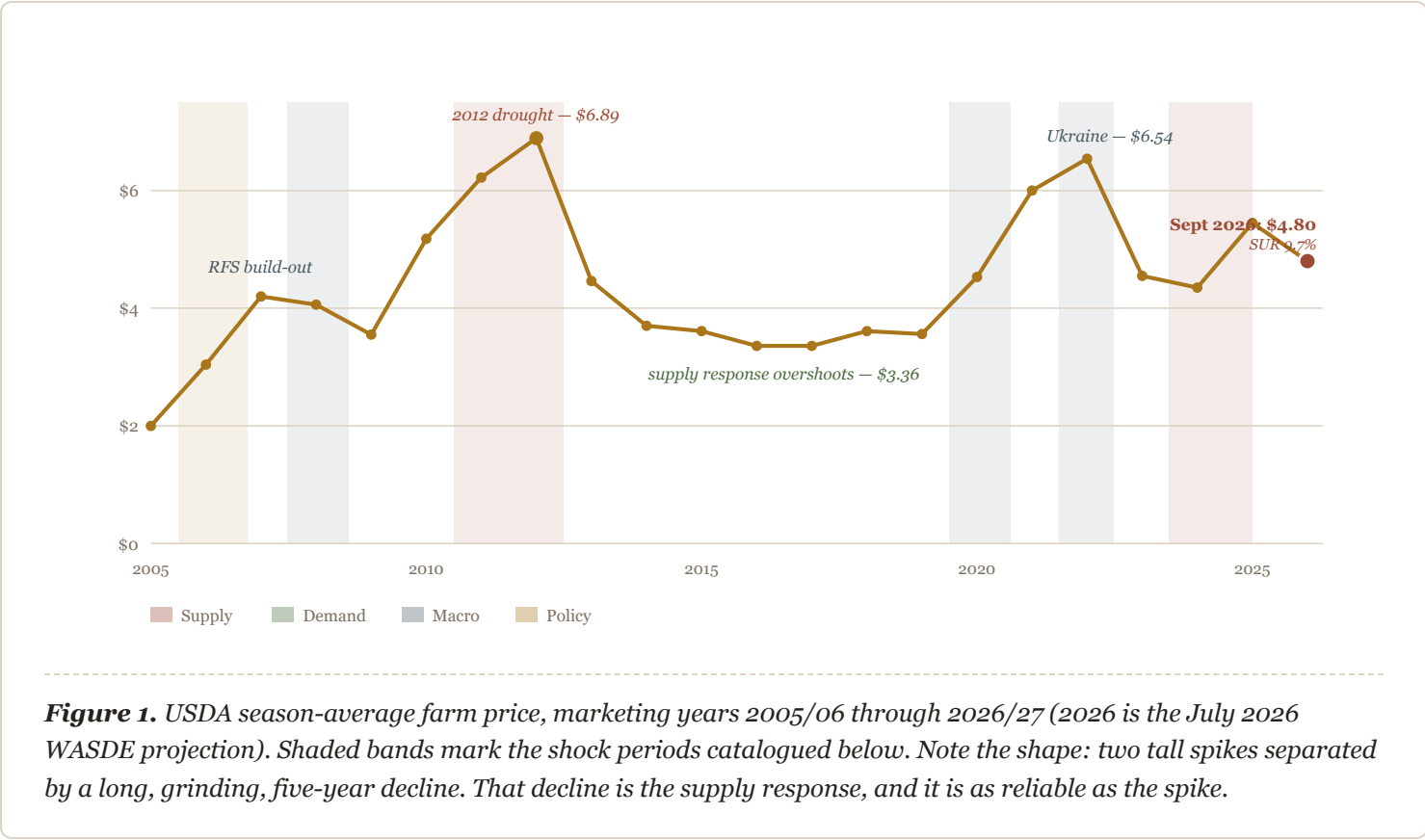


Corn Price Shocks & Preparedness

Twenty years of what moved the market, why, and how a family mill gets ready · Companion to the main analysis · **Revised 25 September 2026**

Corn does not drift. It sits still for years and then moves 60% in six months. Every one of those moves in the past twenty years fits into one of four categories — supply, demand, macro, or policy — and **the category tells you how much warning you get**. That is the practical content of this section. A drought gives you eight weeks. A policy change gives you eighteen months. Only one of those is enough time to go buy grain.

I · TWENTY YEARS OF PRICE



II · THE CHRONOLOGY

PERIOD	TYPE	WHAT HAPPENED, AND WHAT IT DID	WARNING
2005–07	POLICY	The RFS build-out. The Energy Policy Act of 2005 and EISA 2007 created a mandated, price-inelastic demand block where none existed. Season-average price went from \$2.00 (2005/06) to \$4.20 (2007/08). This was not a spike — it was a <i>permanent level shift</i> , and prices never returned to the old base. Roughly a third of the U.S. crop was rerouted into fuel.	18–36 mo
2008	MACRO	Global food crisis and the commodity peak. A weak dollar, \$147 crude, and a broad speculative supercycle carried nearby futures to about \$7.65/bu in June 2008 — a record at the time. The financial crisis then erased it within months. Corn had become a macro asset, not just a feed grain.	3–6 mo
2010–11	DEMAND	Tight stocks meet strong demand. Stocks-to-use fell to roughly 7.9%. Futures matched and then exceeded the 2008 high. Season-average price \$6.22. The convexity of the stocks curve did the work — a small stock decline produced a large price move.	6–12 mo
2012	SUPPLY	The drought. Worst U.S. drought in fifty years. Futures reached roughly \$8.24/bu on 10 August 2012 — the record nominal high, surpassing 2008's \$7.65. Stocks-to-use collapsed to about 7.4%; season-average price \$6.89. The monthly average nearby futures price rose 33% between May and August alone.	6–10 wks
2013–16	SUPPLY	The supply response, and the overshoot. Global acreage surged, yields recovered, and Brazil's <i>safrinha</i> second crop matured into a major export force. Price ground down for five consecutive years to \$3.36 in 2016/17 — below the 2007 level. The single most important pattern in this table: high prices are self-correcting on a two-to-four-year lag, and the correction goes too far.	2–4 yrs
2019	SUPPLY	Prevented planting. A historically wet spring left millions of acres unplanted. A sharp but short-lived rally — a useful reminder that planting-season weather matters as much as pollination weather.	4–8 wks
2020 (spring)	MACRO	COVID demand collapse. Gasoline demand fell off a cliff and took the ethanol grind with it — a third of demand impaired in weeks. Corn fell to the low \$3.00s by April. The fastest demand shock in the modern record, and it arrived with essentially no warning.	< 4 wks

PERIOD	TYPE	WHAT HAPPENED, AND WHAT IT DID	WARNING
2020 (Aug)	SUPPLY	The Iowa derecho. A single storm flattened millions of acres in hours, just as China began an unprecedented buying programme to rebuild its swine herd after African swine fever. Two shocks of opposite sign, four months apart.	0 — none
2021	DEMAND	The post-COVID rally. Chinese import demand, stimulus, freight chaos, and Northern Hemisphere drought. Highest prices since 2013; season-average \$6.00. Notably, China turned to Ukraine for a third of its corn imports — a fact that mattered enormously twelve months later.	6–12 mo
2022	MACRO	Russia invades Ukraine. 24 February 2022. Ukraine supplied roughly 14% of world maize exports . Futures spiked toward the 2012 record within eight weeks; season-average price \$6.54. Research since has found the futures move was partly herding — an overreaction — which is exactly what a buyer who already holds inventory can afford to ignore.	0 — none
2023– 24	SUPPLY	Normalisation. The familiar overshoot again. Gulf FOB corn hit its lowest level since September 2020 in August 2024; national cash corn reached about \$3.87 by late September 2024.	1–2 yrs
2025	SUPPLY	The record crop. 98.7 million acres planted — the most in nearly nine decades — and a record 186.5 bu/acre yield producing 16.8 billion bushels. Stocks-to-use rebuilt to roughly 12.8%.	6–9 mo
2026 (H1)	DEMAND	Export boom, quiet tightening. 2025/26 export sales ran about 31% above the prior year against a USDA target of +12%, on pace for a record, with Mexico taking 18.8 MMT. The July 2026 WASDE cut ending stocks 170 million bushels to 1.80 billion. Price still low, but the direction had changed.	3–6 mo
2026 (Jun– Sep)	SUPPLY	The drought rally — happening now. From a late-June low of \$4.17, September futures ran to a new high of \$5.22 — +\$1.05, or +25%, in nine weeks . Corn and wheat hit their highest levels in more than three years in late August, taking out the late-2023 highs. December settled at \$5.29 on 23 September. Four causes, stacked: nearly 30% of U.S. corn area under moderate-to- exceptional drought since late July, after a mid-July Midwest heat dome and warm August nights; the Pro Farmer Crop Tour scouting 173.2	9 weeks

PERIOD	TYPE	WHAT HAPPENED, AND WHAT IT DID	WARNING
		bu/acre against USDA's 178.5; EU, UK and Brazil production cut on prolonged summer heat during grain fill; and constrained Ukrainian exports pushing U.S. exports <i>up</i> 75 mn bu to 3.3 bn even as production fell.	
		The 11 September WASDE cut yield to 178.5, production to 15.80 bn, and ending stocks to 1.567 bn — stocks-to-use 9.7%, from 11.3% in July. Season-average farm price raised 30¢ to \$4.80.	

This report's own July forecast, graded. The July draft concluded: *"a buying window closing slowly rather than one wide open."* It closed in nine weeks. The framework was right — it named weather as the category with the least warning, put the danger threshold at 9% stocks-to-use, and said the convexity would bite below 10%. The *pacing* was wrong, and in the direction that costs money. Nine weeks is exactly the warning time the chronology assigns to a drought, which is the argument for holding inventory rather than holding a view.

The pattern that should govern policy at the mill. Look down the "warning" column. The two largest price events of the past twenty years — the 2012 drought and the 2022 invasion — gave, respectively, about eight weeks and zero notice. In both cases, by the time the move was visible, the grain was no longer purchasable at the old price. **Below roughly three months of warning, the only thing that protects a mill is grain it already owns.** Everything else — forecasting, hedging, clever contracting — is a slower instrument than the shock.

III • WHY THE CURVE BITES

All of it compresses into one number: the stocks-to-use ratio. The relationship between stocks and price is not a line but a curve, steepening sharply as stocks fall. Fitting the standard reciprocal form to marketing years 2013–2026 gives $\text{Price} = 1.31 + 0.358 \times (1/\text{SUR})$, with $R^2 \approx 0.40$ and a residual standard error of about \$0.80/bu.

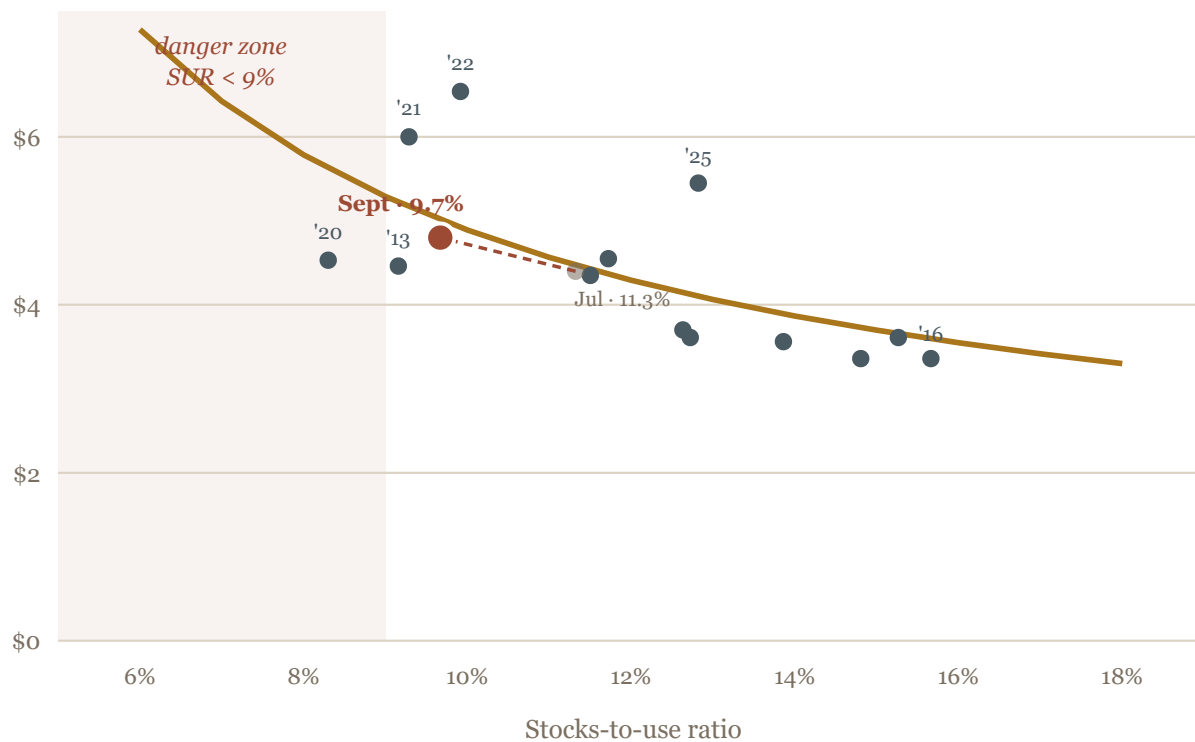


Figure 2. Season-average farm price against stocks-to-use, 2013/14–2026/27, with the fitted reciprocal curve. The 2021, 2022 and 2025 points sit well above the curve — those were the years when export demand and geopolitics, not stocks, set the price. That residual pattern is the honest limit of a one-variable model.

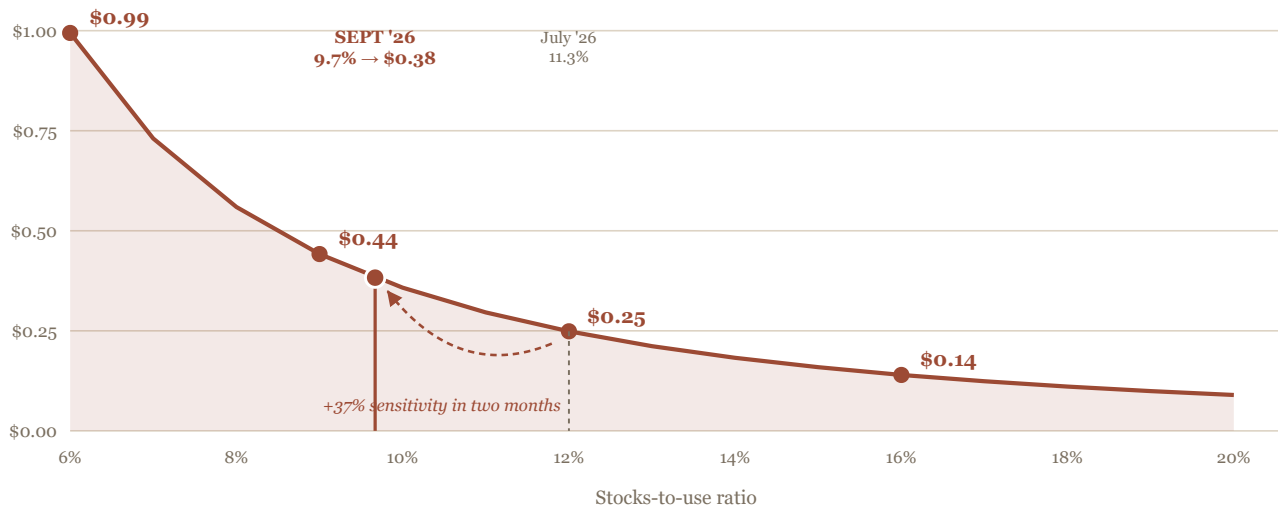
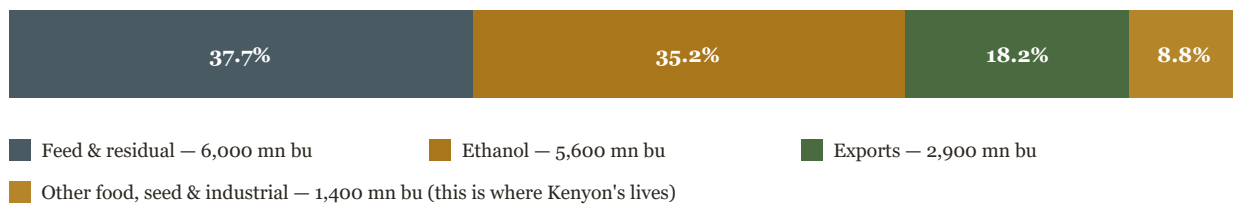


Figure 3. The price impact of a one-percentage-point change in stocks-to-use, computed as $-b/SUR^2$. At 16% stocks a one-point move is worth about 14¢/bu and nobody notices. At 6% the same move is worth 99¢ — seven times as much. This is the entire reason grain markets feel calm for years and then violent for months.

IV • WHAT ACTUALLY CHANGED: THE DEMAND MIX

The RFS did not just raise the price level. It changed the *shape* of the demand curve by inserting a large mandated block that does not respond to price. Ethanol now takes roughly **5.6 billion bushels** a year — about a third of the crop — and USDA's February 2026 baseline holds it flat through 2035.

U.S. corn use, 2026/27 projection — 15,900 million bushels



Mandated and price-inelastic: 35%. Slow-moving: 38%. Volatile and geopolitical: 18%. Price-insensitive and tiny: 9%.

Figure 4. Approximate composition of U.S. corn use. Kenyon's sits in the smallest and least price-sensitive block — a price taker with no market power, but also the segment least damaged by high prices, because pass-through to a premium retail product is high.

V • WHAT A SHOCK ACTUALLY COSTS THE MILL

Here is the number that should reframe the whole question. Take three states of the world — a loose market at 16% stocks, today at 9.7%, and the Crop Tour downside at 6.8% — run the model, apply a 4× flint premium, and convert to a pound of finished meal.

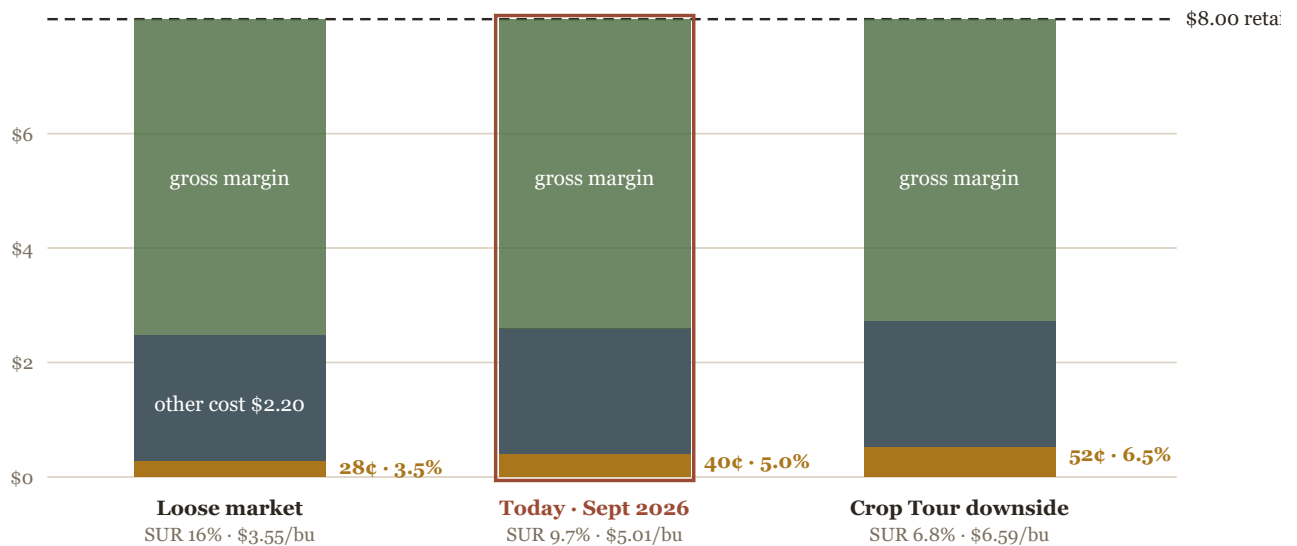


Figure 5. From a loose market to the Crop Tour downside case, flint corn cost per pound of finished meal moves from about 28¢ to 52¢ — a swing of **24¢ on an \$8.00 pound, or 3.0%**. The 25% rally that has already happened since June is worth about **4¢ a pound**, or roughly \$2,100 a year on a 1,000-bushel requirement. A doubling of the corn price still costs the mill less than a single 5% retail price increase recovers.

Read Figure 5 twice. The worst corn shock in fifty years, fully passed through to Kenyon's flint contracts, costs about twenty cents a pound. A 10% reduction in shrink is worth roughly the same. A 5% retail price increase is worth twice as much. This is not an argument for ignoring the corn market — it is an argument for putting it in proportion, and spending management attention where the dollars actually are.

VI • PREPAREDNESS: MATCHING THE RESPONSE TO THE CLOCK

Since the category of shock determines the warning time, and the warning time determines which responses are even physically possible, the preparedness plan writes itself.

SHOCK TYPE	WARNING	WHAT WORKS WITHIN THAT WINDOW
Geopolitical war, trade action, export ban	Days	Only inventory you already hold. Nothing else operates fast enough. This is the entire case for the 12–18 month kernel reserve. Ukraine 2022 gave zero notice and moved the market 30% in eight weeks.
Weather / yield drought, derecho, flood	6–10 weeks	Inventory, plus early warning: NOAA ENSO outlooks quarterly, the U.S. Drought Monitor weekly from June, and NASS Crop Progress condition ratings. A deteriorating July condition rating is the earliest reliable signal you will get. Even so, by the time it is visible the price has moved.
Macro / financial dollar, crude, demand collapse	1–6 months	Watch DXY and WTI monthly. A weakening dollar with rising crude is the classic setup for a corn rally. Enough time to pull forward a scheduled purchase, not enough to build a reserve from nothing.
Export / demand surge China, Mexico, record sales	3–12 months	The weekly FAS Export Sales report on Thursdays is the leading indicator, and it is free. Sales running consistently above USDA's target — as they have all through 2025/26 — means stocks will be revised down. Enough warning to extend contract length and add a grower.
Policy RFS volumes, E15, farm bill	12–36 months	The most warning of any category, and the most consequential. Nationwide E15 would add roughly 2.4 billion bushels of annual demand — a structural shift on the scale of the original RFS. Track EPA volume rules and farm bill progress. Response: lengthen contracts <i>before</i> it happens, and review retail pricing.
Grower loss retirement, isolation failure	Highly variable	This is the mill's real exposure and it is not on anyone's chart. One grower stepping away can remove a large share of a year's White Cap supply, and there is no spot market to replace identity-preserved flint. Response is structural: three to four independent growers on separate isolated parcels, multi-year floor-price contracts, and a seed-stock reserve.

The standing orders

- 1. Hold 12–18 months of whole kernels.** Not futures. Flint kernels store far better than milled meal; store the corn and mill to order. This single action covers the two shock categories that give you no warning, requires no derivatives account, no margin calls, and carries no basis risk. Manage moisture to 13–14% and monitor.

2. **Get to three or four independent growers.** Separate, isolated, geographically dispersed parcels. Highest-priority item in the entire analysis.
3. **Buy commodity grain October–December.** Seasonal lows cluster at harvest; highs cluster in the June–July weather-scare window. Buying in June because you ran out is the most expensive purchasing behaviour available.
4. **Read the WASDE corn table on release day.** Around the 10th–12th monthly, noon Eastern. You need exactly two numbers: ending stocks and total use. Four minutes.
5. **Watch the Thursday FAS export sales number.** The single best free leading indicator of a demand-driven tightening.
6. **Track four internal numbers monthly.** Cost per pound milled, shrink percentage, gross margin by channel, months of flint on hand.
7. **Take small annual price increases.** 3–5% yearly beats an uncomfortable 20% every fifth year, and the customer of a 330-year-old mill is buying provenance, not calories.

Where we are as of 25 September 2026. Stocks-to-use **9.7% and falling**. Model fair value \$5.01/bu; USDA's season-average projection \$4.80; December futures \$5.29. Corn is up 25% off its June low, at a three-year high, and sits in the region where the curve steepens — a further one-point stock decline is now worth **\$0.38/bu**, up 37% from July.

What follows, from the framework rather than from a view on price:

1. The seasonal still says October–December. Harvest supply arrives regardless of the rally. But it will be a higher-priced harvest low than last year's — do not wait past it hoping for 2024 prices.

2. Convexity now outranks percentile. December futures sit at roughly the **79th percentile** of the last fourteen years, which in isolation reads "wait." At 9.7% stocks-to-use with a possible 470-million-bushel yield revision pending, that reading is wrong: the distribution is skewed hard to the upside and the cost of being caught short exceeds the cost of paying up.

3. Lock multi-year flint contracts this autumn. Growers read the same commodity prices you do. At \$5.29 December corn, dent becomes a more attractive alternative for a flint grower — which raises the premium needed to hold acreage. Contracting now is cheaper than

renegotiating in spring, before the 2027 planting decision is made.

4. Keep the cost impact in proportion. The whole move since July is worth about 4¢ a pound. Do not let a 25% headline drive a disproportionate response — but do not use its smallness as an excuse to skip the supply work. The exposure was never the price.

5. Watch three things through October. The 30 September *Grain Stocks* report, the October and November WASDE yield revisions as harvest either confirms or refutes the Crop Tour, and the weekly FAS export sales number. If exports hold at 3.3 billion while yield comes down, stocks-to-use goes below 9% and the curve gets steep fast.

Reproducibility. Every figure here is regenerated from live public data by `build_corn_report.py`, which pulls the macro series from FRED, refits the model, renders the charts with matplotlib, and assembles a formatted Word document. Run it after each WASDE release. The R script `corn_price_model.R` adds the multivariate extension, ARIMA layer, and walk-forward validation against the futures curve.

Data notes. Balance-sheet figures from USDA WASDE and the ERS Feed Grains Yearbook; 2024–2026 are current estimates and projections subject to revision. Futures levels cited are nearby contract highs as reported in the sources listed in the main report. Mill economics in Figure 5 use illustrative operating assumptions — replace them with Kenyon's actual yield, premium, and cost figures in the decision dashboard.